

MA 141 — Test 2 Review Problems

Work these without a calculator. Answers are at the end — try each problem before checking.

1. Differentiate $f(x) = (2x + 1)^4 \cdot \cos x$.
2. Find the equation of the tangent line to $y = \sqrt{x}$ at $x = 9$.
3. A spherical balloon is inflated at $100 \text{ cm}^3/\text{s}$. How fast is the radius increasing when $r = 5 \text{ cm}$?
4. Find the absolute extrema of $f(x) = x^3 - 3x^2 + 1$ on $[-1, 3]$.
5. Verify the Mean Value Theorem for $f(x) = x^2$ on $[1, 3]$ and find c .
6. A farmer has 600 m of fence to enclose a rectangle along a river (no fence on the river side). What dimensions maximize the area?
7. Evaluate $\lim_{x \rightarrow 0} (e^x - 1 - x)/x^2$.

Answers

1. $8(2x + 1)^3 \cos x - (2x + 1)^4 \sin x$
2. $y = (1/6)x + 3/2$
3. $dr/dt = 1/\pi \approx 0.318 \text{ cm/s}$
4. Maximum 1 at $x = 0$ and $x = 3$; minimum -3 at $x = -1$ and $x = 2$
5. $c = 2$
6. $150 \text{ m} \times 300 \text{ m}$ (300 m side parallel to the river), area $45,000 \text{ m}^2$
7. $1/2$